CustomerSegment ML: Automated Customer Segmentation Using K-Means Clustering with Silhouette-Based Model Selection

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Abstract: Customer segmentation lets a business design one marketing strategy per group of similar customers instead of one campaign for everybody. This paper presents CustomerSegment ML, a full-stack system that segments any uploaded customer file automatically. A FastAPI service detects the income and spending columns, removes invalid rows, standardises the features, sweeps the number of clusters K from 2 to 8 and selects the K with the highest silhouette score. K-Means with k-means++ seeding and ten restarts is then fitted, centroids are mapped back to real units and every cluster receives a business persona and a marketing strategy; new customers are classified in real time and the results are explored in a Next.js dashboard. On the 200-customer Mall Customers benchmark the system selects K = 5 and reaches a silhouette score of 0.555, a Davies-Bouldin index of 0.572 and a Calinski-Harabasz index of 248.6. Compared with twelve alternative and baseline methods it ranks first on both silhouette and Calinski-Harabasz, ahead of Gaussian mixtures, hierarchical clustering, BIRCH and DBSCAN, and far ahead of rule-based income tiers (0.131) and random assignment (−0.069). It returned an identical partition on all 30 random seeds tested, while single-run random initialisation fell to 0.347.

Keywords: customer segmentation, K-Means clustering, unsupervised learning, silhouette analysis, elbow method, cluster validation, FastAPI, Next.js

I. Introduction

Retailers record more customer data than ever, yet a single promotion for every customer wastes budget and ignores the obvious fact that customers behave differently. Market segmentation, formalised by Smith [1], divides a market into homogeneous groups that can each be targeted with a suitable strategy. Hand-made rules such as income bands look at one variable at a time and miss the joint structure of behaviour; clustering algorithms discover that structure directly from the data [2].

K-Means [3], [4] is the most widely used clustering algorithm in practice because it is fast, simple and produces centroids that read as the profile of a typical segment member. In practice, however, three problems limit its use by non-experts: the number of clusters K must be chosen in advance, the result depends on the random starting centroids, and the output is a set of numbers rather than an actionable description. Most published work on the popular Mall Customers dataset [5] also stops at a static notebook plot.

This paper presents CustomerSegment ML, a system that addresses these problems end to end. Its contributions are:

- a tolerant upload pipeline that detects columns, cleans invalid rows and standardises features for any customer CSV;
- automatic selection of K by silhouette analysis, supported by the elbow, Davies-Bouldin and Calinski-Harabasz criteria;
- a stable K-Means configuration with k-means++ seeding and ten restarts, validated over 30 random seeds;
- transparent persona and strategy generation and real-time prediction through a REST API and web dashboard;
- a comparison against twelve alternative and baseline clustering methods on identical data.

Section II reviews related work, Section III describes the data, Section IV the method, Section V the implementation, Section VI the results and Section VII concludes.

II. Related Work

A. Clustering Algorithms

Lloyd's algorithm [4] alternates between assigning points to their nearest centroid and moving each centroid to the mean of its points. Arthur and Vassilvitskii proposed k-means++ seeding [6], which chooses spread-out initial centroids and is O(log K)-competitive with the optimum. Mini-Batch K-Means [7] speeds up training with small random batches. Agglomerative clustering with Ward linkage [8] merges clusters with the smallest variance increase, while single, average and complete linkage use other inter-cluster distances. DBSCAN [9] finds dense regions of arbitrary shape, Gaussian mixture models fitted by EM [10] give soft memberships, and BIRCH [11] summarises very large data in a clustering-feature tree.

B. Choosing K and Validating Clusters

Without labels, partitions are judged by internal indices. The elbow method [12] looks for the K after which WCSS stops falling sharply. The silhouette coefficient [13] compares

each point's distance to its own cluster with its distance to the nearest other cluster. The Davies-Bouldin index [14] and the Calinski-Harabasz index [15] measure the ratio of within-cluster scatter to between-cluster separation. The adjusted Rand index [16] compares two partitions and is used here to measure stability across seeds.

C. Segmentation in Marketing

Clustering on demographic and behavioural variables is a standard segmentation method [2], and RFM scoring [17] is a common rule-based alternative. Our work differs from typical studies by packaging the method as a validated service with automatic model selection, explanations and real-time prediction, and by measuring it against a broad set of alternatives.

III. Dataset and Problem Formulation

The system accepts any CSV with an annual income column and a spending column. For evaluation we use the public Mall Customers dataset [5], summarised in Table I. The spending score is assigned by the mall from purchase behaviour; the data have no missing values or duplicates.

TABLE I
Summary of the Evaluation Dataset

Attribute	Type	Range	Clustered
CustomerID	Identifier	1 to 200	No
Gender	Categorical	F 112, M 88	No
Age	Integer	18 to 70	No
Annual income (k$)	Integer	15 to 137	Yes
Spending score	Integer	1 to 99	Yes

Income and spending score are almost uncorrelated ($r = 0.01$) and together show five visible groups, whereas age correlates with spending ($r = -0.33$) and adds no separate structure, so the two behavioural features are clustered and age is kept for profiling. Given n customers $x_i \in \mathbb{R}^2$, the task is to find a partition into K clusters with high cohesion and separation, choose K automatically and assign new customers to the correct cluster.

IV. Proposed Method

A. Preprocessing

Headers are lower-cased and stripped of non-alphanumeric characters so that variants such as “Annual Income (k$)” and “annual_income” are matched. Values are coerced to numbers, rows with missing or infinite features are dropped and reported, and files must contain 10 to 5,000 valid customers. Each feature is then standardised with its mean μ and standard deviation σ so that both contribute equally to the Euclidean distance:

$$z = (x - \mu) / \sigma \tag{1}$$

B. K-Means With k-means++ Seeding

Each customer is assigned to its nearest centroid and each centroid is moved to the mean of its members:

$$c_i = \arg\min_j \| x_i - \mu_j \|^2 \tag{2}$$

$$\mu_j = (1 / |C_j|) \cdot \Sigma\, x_i , \quad x_i \in C_j \tag{3}$$

Both steps decrease the within-cluster sum of squares, which guarantees convergence:

$$J = \Sigma_{j=1}^{K} \Sigma_{i \in C_j} \| x_i - \mu_j \|^2 \tag{4}$$

Initial centroids are drawn with k-means++, where a point is chosen with probability proportional to its squared distance $D(x)^2$ from the nearest seed already selected:

$$P(x) = D(x)^2 / \Sigma_{x'} D(x')^2 \tag{5}$$

The procedure is repeated ten times and the run with the lowest J is kept (random state 42). Centroids are returned to real units by the inverse transform $\hat{x} = \sigma z + \mu$.

C. Automatic Selection of K

For K = 2 to 8 the system fits K-Means and computes the mean silhouette S(K), where a(i) is the mean distance of customer i to its own cluster and b(i) to the nearest other cluster:

$$s(i) = (b(i) - a(i)) / \max\{ a(i), b(i) \} \tag{6}$$

$$K^* = \arg\max S(K), \quad K \in \{2, \ldots, 8\} \tag{7}$$

The Davies-Bouldin (lower is better) and Calinski-Harabasz (higher is better) indices are reported as supporting evidence:

$$DB = (1 / K) \Sigma_{i=1}^{K} \max_{j \neq i} (\sigma_i + \sigma_j) / d(\mu_i, \mu_j) \tag{8}$$

$$CH = [\mathrm{tr}(B_k) / (K - 1)] / [\mathrm{tr}(W_k) / (n - K)] \tag{9}$$

D. Persona Generation and Prediction

A cluster whose mean income and mean spending both meet or exceed the overall means is labelled Premium VIP Shoppers; high income with low spending gives Careful Savers; low income with high spending gives Trendsetters; and the remaining case gives Budget / Value Seekers, each with a matching strategy. A new customer is scaled with the fitted scaler and assigned to the nearest centroid.

V. System Implementation

Fig. 1 shows the architecture. A Next.js 16 dashboard written in TypeScript with Tailwind CSS and Recharts calls a FastAPI service through a typed client. Pydantic schemas validate every request, for example $2 \le K \le 8$, and every model endpoint returns HTTP 409 until a dataset has been uploaded. The segmentation engine, built on scikit-learn [18], fits the scaler and model into local objects and publishes them together, so a concurrent prediction never sees a half-trained model.

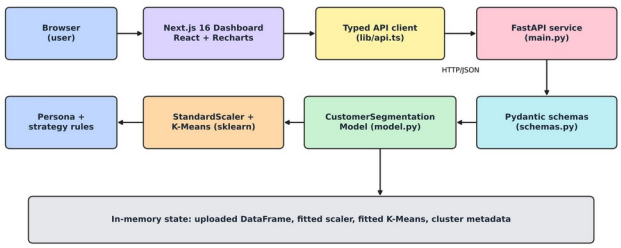


Fig. 1. Architecture of the CustomerSegment ML system.

Six endpoints cover health, upload, dataset, elbow, cluster and predict. The dashboard shows processing stages, the elbow chart, a K slider, the clustered scatter plot with centroids, persona cards, a prediction form and a CSV export of the segmented customers. An automated end-to-end test

uploads a synthetic file with four groups and a broken row and checks every endpoint and error path.

VI. Results and Discussion

A. Selection of K

Table II reports the indices for K = 2 to 8. WCSS drops from 269.7 to 65.6 between K = 2 and K = 5 and then flattens (Fig. 2), the silhouette peaks at 0.555 and Davies-Bouldin reaches its minimum of 0.572, all at K = 5. Three independent criteria therefore agree on five segments.

TABLE II
Validity Indices for K = 2 to 8

K	WCSS	Silhouette	DB	CH
2	269.7	0.321	1.267	95.7
3	157.7	0.467	0.717	151.3
4	108.9	0.494	0.710	174.6
5	**65.6**	**0.555**	**0.572**	**248.7**
6	55.1	0.540	0.655	243.1
7	44.9	0.528	0.715	254.6
8	37.2	0.455	0.760	267.3

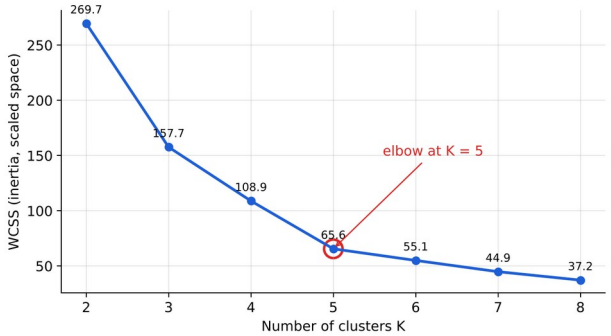


Fig. 2. Elbow curve of WCSS against K.

B. Discovered Segments

Fig. 3 and Table III show the five segments. K-Means converged in four iterations and trained in about 18 ms, so the dashboard can retrain interactively.

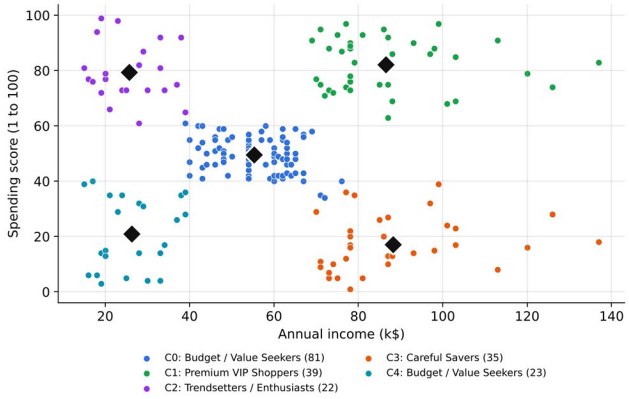


Fig. 3. Final segmentation with K = 5; diamonds mark centroids.

TABLE III
Profiles of the Five Segments

Cluster	Persona	Size	Income	Score
C0	Budget / Value	81	55.3	49.5
C1	Premium VIP Shoppers	39	86.5	82.1
C2	Trendsetters	22	25.7	79.4
C3	Careful Savers	35	88.2	17.1
C4	Budget / Value	23	26.3	20.9

Premium VIP Shoppers (high income, score 82.1) should be retained with loyalty rewards, Careful Savers (income $88.2k, score 17.1) are the largest growth opportunity, Trendsetters are young high spenders with modest income, and two value-seeking groups differ mainly in income. Five hand-made test customers placed in each region were all assigned to the expected segment.

C. Comparison With Alternative and Baseline Methods

All methods were run on the same scaled data with K = 5 (Table IV, Figs. 4 and 5).

TABLE IV
Comparison With Alternative and Baseline Methods (K = 5)

Method	Silh. ↑	DB ↓	CH ↑
CustomerSegment ML	**0.555**	**0.572**	**248.6**
Gaussian mixture	0.554	0.576	244.9
Agglom. (Ward)	0.554	0.578	244.4
Agglom. (complete)	0.553	0.568	247.2
Mini-Batch K-Means	0.539	0.585	241.2
K-Means, random, 1 run	0.502	0.671	201.8
Agglom. (average)	0.479	0.643	143.6
BIRCH	0.414	0.698	125.9
DBSCAN	0.350	0.829	60.8
Agglom. (single)	0.276	0.423	5.6
Income-only K-Means	0.157	2.201	43.1
Income tiers (rule)	0.131	2.529	40.4
Random assignment	-0.069	30.799	0.9

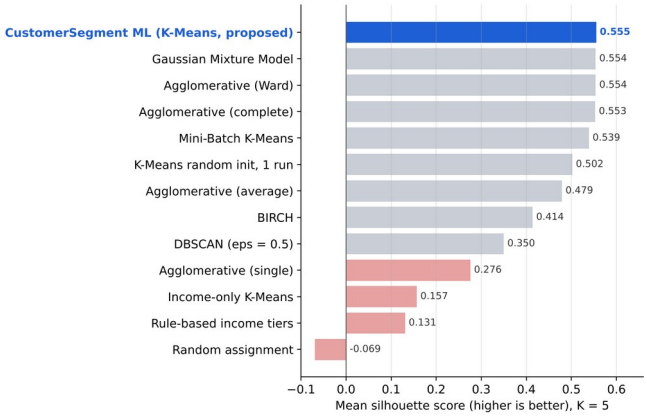


Fig. 4. Silhouette of CustomerSegment ML versus alternative and baseline models.

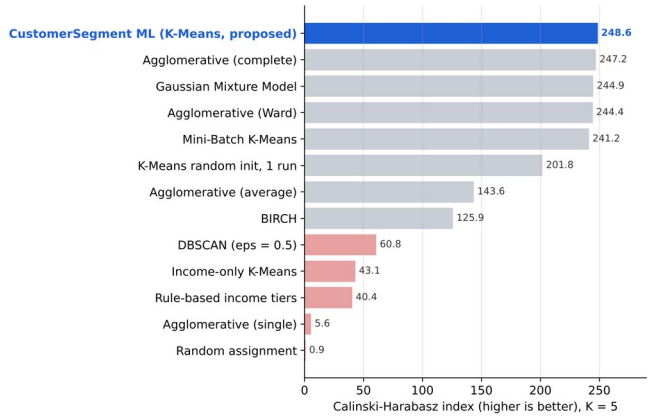


Fig. 5. Calinski-Harabasz index of CustomerSegment ML versus alternative and baseline models.

CustomerSegment ML ranks first on silhouette (0.555) and Calinski-Harabasz (248.6). The Gaussian mixture and Ward clustering come close (0.554) but lack explicit centroids for explanation and instant prediction of new customers. Mini-Batch K-Means loses 0.016, BIRCH 0.141 and DBSCAN 0.205. Clustering on income alone (0.157) or cutting income into tiers (0.131) produces poor segments because it ignores spending behaviour, and random assignment scores −0.069. Single linkage reaches a low DB value (0.423) only by placing almost all customers in one cluster, as its CH of 5.6 shows.

D. Stability

Over 30 random seeds the proposed configuration returned exactly the same partition (silhouette 0.555, standard deviation 0.000, adjusted Rand index 1.000), while K-Means with a single random start averaged 0.503 (standard deviation 0.065) and dropped to 0.347 on an unlucky seed, with only 0.846 agreement between runs (Fig. 6). Reproducibility matters for a business tool, since the same file must always yield the same segments.

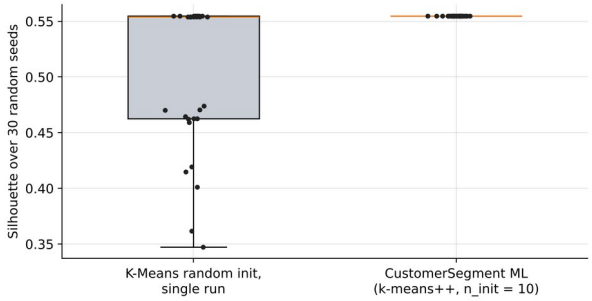


Fig. 6. Silhouette over 30 random seeds.

E. Limitations

The current version clusters two behavioural features, assumes compact clusters, keeps data in server memory and uses rule-based persona names. Internal indices measure geometric quality, not campaign uplift, which requires A/B testing with real customers.

VII. Conclusion and Future Work

CustomerSegment ML turns an uploaded customer file into clear, actionable segments with no manual tuning. Scaling, silhouette-based selection of K and k-means++ with restarts produce five well-separated, perfectly reproducible segments on the Mall Customers benchmark, ranking first among thirteen clustering approaches, while personas, real-time prediction and an interactive dashboard make the result usable by non-experts. Future work will add RFM and product features with dimensionality reduction, let users compare algorithms on their own data, add robust variants such as K-Medoids, persist per-user models and connect segments to campaign tools for uplift measurement.

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